Assignment No: 5

* 1. Count digit in a number

→

```
int n = 54321 , int count = 0;
while (n>0){
    ~~int rem = n%10~~;
        count++;
        n = n/10;
    }
Sop(count);
}
```

o/p → 5

n = 54321		n = 5432
~~n>0~~		5432 > 0 ✓
54321 > 0 ✓		count → 2
count → 1		n = 5432/10
n = 54321/10		→ 543
n = 5432		

n = 543	n = 54	n = 5	n = 0
543 > 0	54 > 0	5 > 0	0 > 0 ✗
count → 3	Cout → 4	Cout → 5	
n = 543/10	n = 54/10	n = 5/10	
→ 54	→ 5	→ 0	

100/10 → 10 , 999/10 = 99 , 54321/10 → 5432

2) Extract all digits

```
int n = 5382
while (n>0){
    int lastdigit = n%10;
        Sop(lastdigit);
        n = n/10;
    }
```

o/p → 2
8
3
5

n%10 ⇒ give last digit of n

n = 5382	n = 53
n > 0 ✓	53 > 0
Ld = 5382%10	LD = 53%10 ⇒ 3
→ 2	n = 53/10 ⇒ 5
n = 5382/10	n = 5
→ 538	n > 0 ✓
n = 538	LD = 5%10 = 5
n > 0	n = 5/10 → 0
LD = 538%10	n = 0.
→ 8	0 > 0 ✗
n = 538/10	
n = 53	

```
* int ~~sum~~ n = 543
  int sum = 0;
  while (n>0){
    int rem = n%10;
      sum = sum + rem;
      n = n/10;
    }
  Sop(sum);
}
```

o/p → 12

5 + 4 + 3 = 12

* Armstrong number

n = 153 , count = 0.

153 % 10 → 3 ; 15 % 10 → 5 ; 1 % 10 → 1

count → 1 ; count → 2 ; count → 3

153 / 10 → 15 ; 15 / 10 → 1 ; 1 / 10 → 0.

n = 153

rem = 153 % 10 → 3 ; ~~int result = 1~~ ; rem = 5 ; 15 % 10 → 5

~~int rem~~ ~~sum = sum + 3~~

```
for(int i=1; i<count; i++){
    ~~sum = sum +~~
    ~~result = result * count~~;
    rem = rem * count;
}
```

rem = 5 × 3 = 25

25 × 3 = 125.

27 → sum = sum + rem; —— 27 + 125 = 152.

n = 153/10 → 15. 15/10 → 1.

1 % 10 → 1

1 × 3 = 1 152 + 1 = 153

1 × 3 = 1

Q.4 Palindrome Number

```
int n = 121 , int rev = 0;
int dup = n;
while(dup>0){
  int rem = dup%10;
  rev = rev*10+rem;
  dup = dup/10;
}
if(n == rev){
    sop(palidrome);
}
else{ sop(Not Palidrome);
}
```

dup = 121

dup % 10 → 1

rev = 0*10 + 1

= 1

dup / 10 → 12

dup → 12

rem = 12 % 10 → 2

rev = 1*10 + 2

= 12

dup / 10 → 1

dup → 1.

rem = 1 % 10 = 1

rev = 12 * 10 + 1

= 121

dup / 10 = 1 / 10 → 0

dup = 0

0 > 0 X

if(n == rev) =

121 == 121 → Palidrome.

123 = n
321 = rev } if(n == rev) X
else{ Not Palidrome

Q5) • sum; digits

rem = n % 10 = add = add + rem.

n = 543

543 % 10 → 3

add = 0 + 3 = 3

543 / 10 → 54

54 % 10 = 4

add = 3 + 4 = 7

54 / 10 → 5

5 % 10 → 5

add = 7 + 5 = 12

5 / 10 → 0

* Reverse number

~~int n = 12345, int rev = 0;~~
~~while (n>0)~~

12345 ~~rev = rev + 5 = 5~~

~~rem~~ = n%10 → 5

1234%10 → 4

rev = rev * 10 + rem
= 0 * 10 + 5 = 5
= 5 * 10 + 4 = 54
= 54 * 10 + 3 = 543

```
int n = 12345
int reverse = 0;
while(n>0){

int lastdigit = n%10;
reverse = reverse * 10 + lastdigit;

n = n/10;
}
Sop(reverse);
```

o/p → 54321

n = 12345
12345 > 0
↓
LD = 12345%10 → 5
rev = 0 * 10 + 5 ⇒ 5
n = 12345/10
n = 1234
1234 > 0
LD = 1234%10 ⇒ 4
rev = 5 * 10 + 4
= 50 + 4 ⇒ 54
n = 12345/10
n = 123

n = 123
123 > 0
↓
LD = 123%10 ⇒ 3
rev = 54 * 10 + 3
= 543
n = 123/10 ⇒ 12

n = 12
12 > 0
LD = 12%10 ⇒ 2
rev = 543 * 10 + 2
= 5432
n = 12/10 ⇒ 1

n = 1
1 > 0
LD = 1%10 → 1
rev = 5432 * 10 + 1
= 54321
n = 1/10 ⇒ 0

n = 0
0 > 0 X

```
int n = 12
    int count = 0;
for(int i=1; i<= math.sqrt(n); i++){
        if( n%i ) == 0){
                sop(i); count++;
        if( i != n/i){
            sop(n/i);
            count++;
        }
    }
sop(count);
}
```

n = 12
divisors 1, 2, 3, 4, 6, 12
o/p → 6

Eg. prime number check

7 → 7/1 = 0 7/7 = 0 — 2
9 ⇒ 9/1 = 0 9/3 = 0 9/9 → 0 — 3
10 ⇒ 10/1 = 0 10/5 = 0 10/10 → 0 — 3
5 → 5/1 = 0 5/5 = 0 — 2.

count 2 prime
count 3 and more — Not prime

int n = 3, int count = 0;
~~for in~~
for loop → i=1 1<=3 i=2 2<=3 i=3 3<=3
if(n%i == 0) ✓ n%i == 0 ✗ n%i == 0 ✓
1 2 2

int n = 4
for loop i=1 i=2 i=3 i=4
4%1 == 0 ✓ 4%2 == 0 ✓ 4%3 == 0 ✗ 4%4 == 0 ✓
1 2 3

* Print all divisors.

n = 12

12/1 → 12 ✓ 12 1 12

12/2 → 6 ✓ 12 2 6

12/3 → 4 ✓ 12 3 4

12/4 → 3 ✓

12/5 → ✗

12/6 → 2

12/7 →

```
int n = 12
for(int i = 1; i<= math.sqrt(n); i++){
    if( n%i == 0){
        sop(i)

        if (i != n/i){
            sop(n/i)
        }
```

o/p →
1 2 3 4 6 12

i=1 1<=3
↓
12/1 == 0 ✓
1 Count
1! == 12 ✓
↓ 12

i=2 2<=3
12/2 == 0 ✓
2
2! == 6 ✓
6

i=3 3<=3
12/3 == 0 ✓
3
3! == 4
4

* Count Number of Divisors

i=1 1<=3	i=2 2<=3	i=3 3<=3
12/1 == 0 ✓	12/2 == 0	12/3 == 0 ✓
1 Count → 1	2 Count → 3	3 Count → 5
1! == 12 ✓	2! == 6 ✓	3! == 4
↓	6	4
12 Count → 2	Count → 4	Count → 6

```
int n= 153 ; int copy = n;
int dup= n;
int count = 0; int sum = 0;
 while (dup > 0){
   ~~int rem~~=
     ~~n = n/10~~; dup = dup/10;
     count++;
     }

while ( n>0){
 int rem = n%10;

for(int i= 1; i<count; i++){
    ~~sum = s~~
    rem = rem * count;
    }
  sum = sum + rem;
    n = n/10;
    }
  ~~sop(sum)~~;      o/p -> 153
  }
if( copy == sum){
     sop( Armstrong number).
  }
else {
     sop( ~~No Armstrong number~~)
     }.
```

```
while (n>0)
int rem = n%10;
sum = sum + Math.pow
```

```
int a = 8;
int b = 15;
int gcd = 1;
for(int i=1; i<= Math.min(a,b); i++){
    if(a%i == 0 && b%i == 0){
        ~~gcd++~~ gcd = 1;
    }
}
if (gcd == 1){
    sop( coprime number);
}
else{ sop( No coprime number);
}
}
```

o/p/ → coprime number

```
* int n=6
  int sum= 0;

  for(int i=1; i<n; i++){
      if(n%i == 0) {
          sum = sum + i;
      }
  }
  ~~Sop (sum~~
  if( sum == n){
      sop( Perfect number);
  }
  else sop( Not perfnumber);
```

i=1
6/1 = 0 ✓
sum = 1

i=2 6/2 = 0 ✓
sum = 1+2 = 3

i=3 6/3 =
sum = 3+3 = 6

i=4 6/4 = 2 ✗

i=5 6/5 == 0 ✗

~~sop~~

sum=6 n=6

Perfect number

Q 11

a=12 — 1 2 3 4 6 12
b=18 1 2 3 6 9 18

i=1 i<= Math.Min(12,18);
12

12=12

12%1 == 0 && 18%1 == 0 gcd = 1

12%2 == 0 && 18%2 == 0 gcd = 2

12%3 == 0 18%3 == 0 gcd = 3

12%4 == 0 && 18%4 == 0 ✗

12%5 == 0 ✗

12%6 == 0 && 18%6 == 0 gcd 6

12%7 == 0 ✗

12%12 == 0 && 18%12 == 0 ✗

Code

```
int a=12
int b=18
int gcd=1;
for(int i=1; i<= Math.min(a,b); i++){
        if(a%i == 0 && b%i == 0){
            gcd=i;
        }
    }
sop(gcd);
```

o/p → 6

Q.13 Coprime numbers

8 — 1, 2, 4, 8
15 — 1, 3, 5, 15

जिन दोनों का divisor fact 1 है। वो same अपवाद.

gcd=1 gcd>=1

Code ⟹

```
int n= 7
int count = 0;
for(int i=1 ; i<=n ; i++){
    if(n%i==0){
        count++;
    }
}
if(count==2){
    sop(prime);
}
else {
    sop(notprime);
}
```

i=1, i=7

7/1=0 7/7=0

count==2 prime

op ⇒ prime

Q.10 Print All prime Number up to N

```
int n=20
int count
for(int i=1; i<=n; i++){
    int count = 0;
    for(int j=1 ; j<=n; j++){
        if(i%j==0){
            count++;
        }
    }
    if(count>2){
        sop(i
    if(count<=2){
        sop(i+" ");
    }
```

n=3

i=1 1<=3

c-0

j=1 j<=3

1%1==0

c-1

2%1==0

c-2

3%1==0 c-3 → 3<=1

j=2 2<=3

c=0

j=1 1%2==0 c-1

2%2== c-2

3%2==0 ✗ 2<=2 ✓

2

i=3 3<=3

c=0

j=1 1%3==0 c-1

2%3==0 ✗

3%3==0 ✓ c=2

2<=2 ✓

3

1 to 3 → Prime number 2 3.

Q.13

Q 11

a=1

b=1

Code ⟹